FACULTY OF ENGINEERING AND TECHNOLOGY

HOCHSCHULE FURTWANGEN UNIVERSITY HFU

Master Thesis

Fluid 4.0-Compliant Framework for Digital Representation and Graph-Based Neural Modeling

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Abstract

Fluid 4.0 was conceived as a standardized digitalization framework for industrial hydraulics, enabling machine-readable, vendor-independent descriptions of hydraulic components and systems through the Asset Administration Shell (AAS). While existing Fluid 4.0 submodels primarily focus on identification, documentation, and operational data, they do not explicitly represent physical degradation mechanisms required for explainable condition monitoring and predictive maintenance.

This thesis extends the Fluid 4.0 concept by introducing a dedicated submodel for *degradation relationships*, in which physical degradation mechanisms are formally described using causal graph structures. The proposed submodel encapsulates domain knowledge about component wear, operating conditions, and physical constraints, allowing degradation to be represented as structured, interpretable information within the AAS rather than as implicit correlations learned solely from data.

Two complementary study cases are presented to analyze explainability in hydraulic condition monitoring. The first study demonstrates causal-graph-based explainability for an axial piston pump, where geometric wear is characterized through a normalized degradation coefficient α. Unlike conventional volumetric efficiency metrics, which are strongly influenced by operating conditions, the α-parameter captures geometric degradation by explicitly accounting for pressure and viscosity effects. The degradation behavior is analyzed under distinct machine states, resulting in multiple degradation trends that are evaluated independently and subsequently combined into a physically interpretable health indicator. These relationships are encoded in a causal degradation graph and integrated into the Fluid 4.0 AAS.

The second study investigates accelerated degradation in a hydraulic cylinder operating without an active cooling system. This case emphasizes event-based explainability using operational data and user-defined inputs, illustrating the limitations of purely data-driven explanation methods when physical context is weak. A comparative analysis highlights that graph-based, physics-informed explanations provide more consistent and trustworthy interpretations by directly linking observed anomalies to physical degradation states rather than statistical correlations alone.

Building on the defined causal graphs, this thesis proposes a graph neural network (GNN) training strategy that leverages the degradation submodel as structured prior knowledge. While full industrial integration and large-scale deployment are identified as future work, the results demonstrate that physics-informed, graph-based representations significantly improve interpretability, robustness, and user trust in condition monitoring outcomes.

Overall, this work establishes a practical foundation for embedding physical degradation knowledge into Fluid 4.0-compliant digital twins, showing that explainable, physically grounded models are essential for the acceptance and effective use of predictive maintenance solutions in industrial hydraulic systems.